

Law of multiplication

$$P(X \cap Y) = P(X) \cdot P(Y|X) = P(Y) \cdot P(X|Y)$$

If X & Y are independent events,

$$P(X) = P(X|Y) \text{ \& } P(Y) = P(Y|X)$$

consequently,

$$P(X \cap Y) = P(X) \cdot P(Y)$$

Law of addition

$$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$$

Law of conditional probability

$$P(X|Y) = \frac{P(X \cap Y)}{P(Y)} = \frac{P(X) \cdot P(Y|X)}{P(Y)}$$

↓ joint probability ↓ Marginal probability

Bayes' rule

$$P(X|Y) = P(Y|X) P(X)$$

probability distributions

Expected value of X .

Let X be a discrete random variable with set of possible values D & pmf $p(x)$. The expected value or mean value of X , denoted.

$E(X)$ or μ_X is

$$E(X) = \mu_X = \sum_{x \in D} x \cdot p(x)$$

Variance & Standard deviation

Let X have pmf $p(x)$ & expected value μ

$$\begin{aligned} V(X) = \sigma^2 &= \sigma_X^2 = \sum_D (x - \mu)^2 \cdot p(x) \\ &= E[(X - \mu)^2] \end{aligned}$$

$$SD = \sigma_X = \sqrt{\sigma_X^2}$$

$$\begin{aligned} V(X) = \sigma^2 &= \left[\sum_D x^2 \cdot p(x) \right] - \mu^2 \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

Mean of a discrete distribution

$$\mu = E(X) = \sum x \cdot p(x)$$

Covariance

Covariance betⁿ 2 random variables can be +ve or -ve.

If 2 random variables move in the same direction then the covariance will be +ve, if they move in opposite direction, the covariance will be -ve.

Properties

If X & Y are independent random variables, then covariance is zero

$$E(XY) = E(X) \cdot E(Y)$$

$$\text{cov}(X, Y) = 0$$

Correlation

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

Some special distributions

Discrete

- Binomial
- Poisson
- Hypergeometric

Continuous

- uniform
- Exponential
- Normal

Binomial Distribution

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

$$\text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

$$\text{correlation coefficient} = \frac{\text{cov}(x, y)}{s_x s_y}$$

$$m = \underset{\substack{\text{slope of} \\ \text{regression} \\ \text{eqn.}}}{\text{cov}(x, y)} \over \text{var } x$$

Binomial Distribution

$$\text{probability fn} = P(X) = \frac{n!}{x!(n-x)!} p^x q^{n-x} \quad \text{for } 0 \leq x \leq n$$

$$\text{Mean value } \mu = n \cdot p$$

$$\text{variance } \sigma^2 = n \cdot p \cdot q$$

$$\text{std deviation } \sigma = \sqrt{\sigma^2} = \sqrt{n \cdot p \cdot q}$$

Poisson distribution

$$P(X) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$\begin{aligned} \lambda &= \text{long run average} \\ \text{mean} &= \text{variance} = \lambda \\ \text{std deviation} &= \sqrt{\lambda} \end{aligned}$$

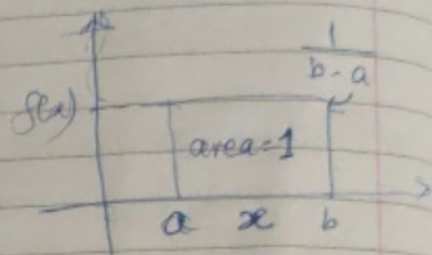
Hypergeometric distribution finite

Continuous probability distributions

Uniform
Exponential
Normal

Uniform distribution

$$f(x) = \frac{1}{b-a} \text{ for } a \leq x \leq b$$
$$= 0 \text{ for all other values}$$



The uniform distribution is a probability distribution that has equal probabilities

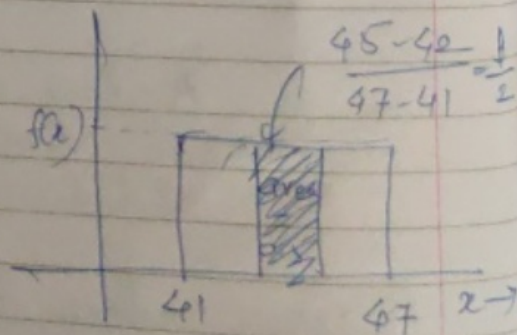
Example: Uniform probability distribution over the range $2 \leq x \leq 6$

$$f(x) = \frac{1}{6-2} = .25 \text{ for } 2 \leq x \leq 6$$

$$\mu = \frac{a+b}{2} = \frac{2+6}{2} = 4$$

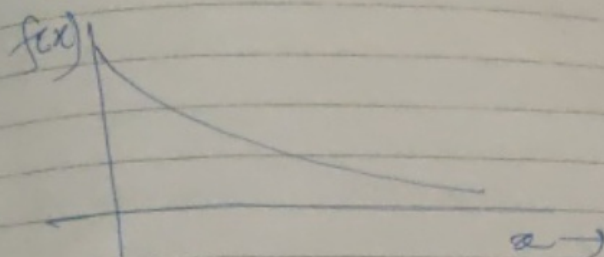
$$\sigma = \sqrt{\frac{(b-a)^2}{12}} = \sqrt{\frac{6-2}{12}}$$

$$P(x_1 \leq x \leq x_2) = \frac{x_2 - x_1}{b - a}$$



Exponential probability distribution

$$f(x) = \frac{1}{\mu} e^{-x/\mu}$$



Normal Distribution

bell shaped

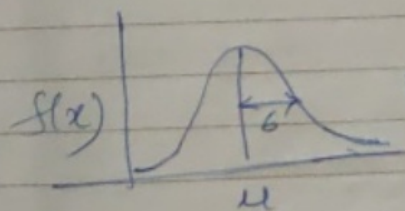
symmetrical

Mean, mode, median are equal.

location is characterized by μ ,

Spread is " " std deviation σ

Random variable has $-\infty$ to $+\infty$



Mean = mode = median

PDF

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$e \approx 2.71828, \pi \approx 3.14159$$

μ - population mean, σ - standard deviation of population

Standard Normal Distribution

It has mean = 0, std deviation = 1

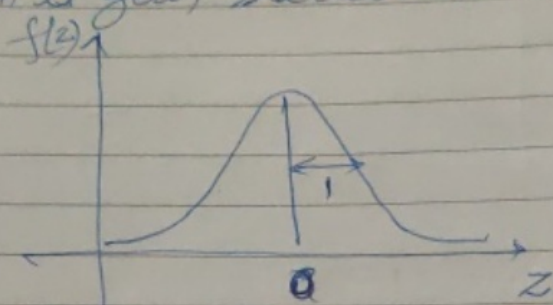
Any normal distribution can be converted to std. normal distribution by using z distribution

$$Z = \frac{X - \mu}{\sigma}$$

standardized normal prob. d f.

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

Standardized normal distribution also called as z-distribution
mean is zero, std deviation is 1



Week 3 - Sample variance

Let $x_1, x_2, \dots, x_n$ be a random sample from a population.
The sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Square root of s^2 is ^{sample} standard deviation

chi-square example

A commercial freezer must hold a selected temperature with little variation. Specifications call for a std deviation of no more than 4 degrees.

A sample of 16 freezers is to be tested. What is the upper limit (K) for the sample variance such that the probability of exceeding this limit, given that the population std deviation is 4, less than .05

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

$\bar{x}$ - sample mean

s^2 - " variance

p - " proportion.

General formula for all confidence interval is

$$\text{point estimate} \pm (\text{Reliability factor})(\text{std error})$$

The value of reliability factor depends on the desired level of confidence

Confidence interval for μ (σ^2 known)

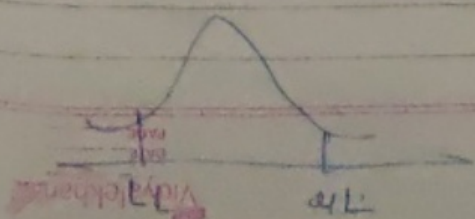
- population variance σ^2 is known

- " is normally distributed

- If population is not normal, use large sample

Confidence interval estimate:-

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$



$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Margin of error.

$$\bar{x} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$ME = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

σ - error
$\frac{\sigma}{\sqrt{n}}$ std error.
$Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ - margin of error.

confidence level	$Z_{\alpha/2}$ value
90 %	1.645
95 %	1.96
99 %	2.58

If σ^2 is unknown, how to predict population mean.

- Consider a random sample of n observations with mean $\bar{x}$ & standard deviation s .
- from a normally distributed population with mean μ
- Then the variable $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

follows the student's t distribution with $(n-1)$ degrees of freedom.

Confidence interval estimate

$$\bar{x} - t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} < \bar{x} + t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$$

$$t_{n-1, \alpha/2} = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Here ME (margin of error) $= t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$

z value depends on degrees of freedom
 $df = n - 1$

Hypothesis Testing

Three approaches for Hypothesis Testing

- 1) P-value
- 2) Critical value
- 3) Confidence Interval value

1) P-value says whether it supports null hypothesis or not
p-value is the probability using the test statistic that measure the support provided by the sample for null hypothesis

2) If p-value is $<$ or $=$ to level of significance α , the value of test statistic is in the rejection region

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \quad t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Reject H_0 if p-value $\leq \alpha$

Steps of Hypothesis testing - P-value approach

Step 1 - Develop the null & alternative hypothesis

2 - Specify the level of significance α

3 - collect the sample data & compute the test statistic (t, Z)
p-value approach

Step 4 - Also the value of test statistic to compute p-value

Step 5 - Reject H_0 if p-value $< \alpha$

Week 3 - Sampling Distribution

If the population is normal with mean μ & std deviation σ , the sampling distribution of $\bar{X}$ is normally distributed with

$$\mu_{\bar{x}} = \mu \quad \text{and} \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \quad \text{standard error of mean}$$

If the sample size 'n' is not large relative to the population size N, then

$$\mu_{\bar{x}} = \mu \quad \text{and} \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

Z-value for sampling distribution of mean

$$Z = \frac{\bar{X} - \mu}{\sigma_{\bar{x}}}$$

↓
std error of mean

Central limit theorem

If the population is not normal, sample means from the population will be approximately normal as long as sample size is large enough

$$\mu_{\bar{x}} = \mu, \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

$\sigma \rightarrow$ population std dev
 $n \rightarrow$ sample size

Sampling distributions of sample proportions
sample proportion ($\hat{p}$) provides an estimate of p

$$\hat{p} = \frac{x}{n}$$

$$0 \leq \hat{p} \leq 1$$

$$E(\hat{p}) = p - \text{mean}$$

$$\sigma_{\hat{p}}^2 = \text{Var}\left(\frac{X}{n}\right) = \frac{p(1-p)}{n}$$

In Binomial distribution
 $np = np$, $\sigma_{\hat{p}}^2 = npq$

Z-value for proportion

$$Z = \frac{\hat{p} - p}{\sigma_{\hat{p}}} = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

$\hat{p}$ - sample proportion

p - Population proportion

Sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

sample standard deviation $= \sqrt{s^2} = s$

Sampling distribution of s^2 has mean σ^2
 $E(s^2) = \sigma^2$

if the population distribution is normal
 $\frac{(n-1)s^2}{\sigma^2}$ has χ^2 distribution with $n-1$
degrees of freedom.

Week 4

2-sample test

$$\text{var}(A-B) = \text{var}(A) + \text{var}(B)$$